

# ADNAN MASHRUR SADAD

## AI / ML ENGINEER | APPLIED AI & ML SYSTEMS

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GitHub | LinkedIn | Portfolio

Software engineering graduate building LLM applications and deployable ML pipelines, with enterprise development experience at Joget and first-author research in RAG and Text-to-SQL evaluation.

### Technical Skills

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**AI & Data:** Python, SQL, PyTorch, scikit-learn, XGBoost, pandas, NumPy, Hugging Face, RAG, LLM evaluation

**Applications:** FastAPI, Pydantic, React, TypeScript, Groq, Text-to-SQL, structured-output validation

**ML Systems:** Docker, Google Cloud, Kafka/Redpanda, PyFlink, MLflow, ONNX, GitHub Actions

### Experience

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**Software Developer Intern** | *Joget Inc.*

**Oct 2024 - Mar 2025**

- Built and deployed **Joget DX workflow automations and UI components** for client teams, working with senior engineers, QA, and product managers in Agile sprints.
- Integrated **REST APIs** to remove manual data re-entry and maintained application modules through recurring production release cycles.

### Selected Projects

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**Driftline** | Streaming Fraud Detection and MLOps | GitHub

**2026**

- Built a streaming fraud-detection pipeline replaying **590K transactions** through Redpanda/PyFlink; engineered a GraphSAGE model over a **605K-node identity graph** with unit-tested controls against data leakage.
- Evaluated XGBoost using a chronological split: **0.4751 PR-AUC** versus 0.6974 on a random split. Implemented drift-triggered retraining with a promotion gate; candidate PR-AUC was **0.4802 versus 0.2811** for the incumbent on the same holdout.
- Deployed XGBoost inference through **ONNX and FastAPI on Google Cloud**; load-tested 10,018 requests with zero failures and **380 ms p95 latency at 50 concurrent users** on a 4-vCPU VM.

**InterviewPilot** | Applied AI Interview Platform | GitHub

**2026**

- Built a **React/FastAPI** interview application with persistent sessions, Groq transcription, schema-validated AI answer evaluation, and bounded agentic follow-ups; added repair loops and disclosed fallback scoring.
- Validated the backend with **19 tests and 89% statement coverage**, plus CI browser checks from session creation to report generation; external AI calls were mocked or disabled in tests.

**Freight Rate ML** | Predictive Modelling | GitHub

**2026**

- Benchmarked regression models on a **9,523-row temporal holdout** with training-only preprocessing; boosted MAPE was **6.89% versus 8.28%**, while the linear baseline retained better MAE/RMSE.

### Research

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**SAFE-RAG** | Regulatory RAG Evaluation | GitHub

**2025 - 2026**

- First author; **short paper under review at ALTA 2026**. Evaluated answers with valid output schemas and supporting citations; identified an **8.6% wrong-scope grounding rate** across 200 annotated items.

**Text-to-SQL Answerability under Schema Evolution**

**2026**

- First author; manuscript complete, not yet submitted. Built a **1,382-instance benchmark across 20 databases** to evaluate answerability as database schemas change; ran 27,384 model generations.
- Detected **79.5% of structurally invisible drift at a 13.0% false-alarm rate**; evaluated against 5,627 benign updates.

### Education

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**Universiti Teknologi Malaysia (UTM), MJIIT** | Kuala Lumpur | 2021 - 2026

B.Sc. Computer Science (Software Engineering) with Honours | **CGPA: 3.43/4.00**

Dean's List: 2022/2023, 2025/2026.

### Awards & Leadership

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**Gold Medal & Best Video Award**, myHCI-UX Student Design Challenge (Oct 2025).

**Secretary Treasurer, SOFEA Society** (2022 - 2023); organised AI workshops for 200+ students.